Seminar Notes on Kodaira's *Complex Manifolds and Deformation of Complex Structures, I*

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Preface

These notes were written for a reading seminar on K. Kodaira, *Complex Manifolds and Deformation of Complex Structures, I* (1968). Their main aim is to fill in, by hand, the derivations and details that the book omits.

Principles for these seminar notes

1. Prove what is written in the text; reconstruct each claim in one's own words.
2. Abbreviate notation as convenient.
3. Prioritize understanding: explain every point that is omitted or glossed over.
4. Reconstruct proofs oneself, or revise the flow of the argument as needed.
5. Write down as many questions and supplementary remarks as possible (kept separately from the preparation notes).
6. Prepare presentation notes (so that one can present without consulting the preparation).

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Chapter 1

Holomorphic Functions

1.1 Holomorphic functions of several variables

Notation and preliminaries

Let the n-dimensional complex space be

$$\mathbb{C}^n = \{\, z = (z_1, \dots, z_n) \mid z_\nu \in \mathbb{C} \,\},$$

and write each coordinate as $z_\nu = x_\nu + iy_\nu$ ($x_\nu, y_\nu \in \mathbb{R}$, $i = \sqrt{-1}$). Through the correspondence $z \mapsto (x_1, y_1, \dots, x_n, y_n)$ taking real and imaginary parts, we identify $\mathbb{C}^n \cong \mathbb{R}^{2n}$.

Define the norm of $z = (z_1, \dots, z_n)$ by

$$|z| = \Bigl(\sum_{\nu=1}^{n} |z_\nu|^2\Bigr)^{1/2}.$$

For $z, w \in \mathbb{C}^n$ the following hold:

$$|z_\nu| \le |z| \le \sum_{\nu=1}^{n} |z_\nu|, \tag{1.1}$$

$$|z + w| \le |z| + |w|, \tag{1.2}$$

$$\bigl|\, |z| - |w| \,\bigr| \le |z - w|. \tag{1.3}$$

Holomorphic functions

Def 1.1. Let $D \subseteq \mathbb{C}^n$. We say that $f(z) = f(z_1, \dots, z_n)$ is a *holomorphic function* on D if f is continuous on D and is holomorphic in each variable z_ν separately (the other variables being held fixed).

Domains and polydiscs

Def 1.2 (domain)**.** A connected open subset of $\mathbb{C}^n$ is called a *domain*.

Note 1.1. A topological space X is *connected* if every decomposition $X = U \cup V$ with $U \cap V = \varnothing$ and U, V open satisfies $U = \varnothing$ or $V = \varnothing$ (that is, X cannot be split into two nonempty open sets). For open subsets of $\mathbb{C}^n$, connectedness is equivalent to path-connectedness (any two points can be joined by a path within the set).

Def 1.3 (polydisc)**.** For $a = (a_1, \dots, a_n) \in \mathbb{C}^n$ and $r = (r_1, \dots, r_n)$ ($r_\nu > 0$),

$$P(a, r) = \{\, z \in \mathbb{C}^n \mid |z_\nu - a_\nu| < r_\nu,\ \nu = 1, \dots, n \,\}$$

is called the *polydisc* with center a and radius r.

Note 1.2. With the maximum norm $\|z\| = \max_{1 \le \nu \le n} |z_\nu|$, a polydisc is the "ball" $\{\, z \mid \|z - a\| < r \,\}$ with respect to $\|z - a\|$ (when $r_1 = \cdots = r_n = r$).

Wirtinger derivatives (complex derivatives) and the Cauchy–Riemann equations

For $z_\nu = x_\nu + iy_\nu$, define the *Wirtinger derivatives* by

$$\frac{\partial}{\partial z_\nu} = \frac{1}{2}\Big(\frac{\partial}{\partial x_\nu} - i\frac{\partial}{\partial y_\nu}\Big), \qquad \frac{\partial}{\partial \bar{z}_\nu} = \frac{1}{2}\Big(\frac{\partial}{\partial x_\nu} + i\frac{\partial}{\partial y_\nu}\Big).$$

In what follows we split a C^1 function f into real and imaginary parts, $f = u + iv$ (u, v real-valued).

Def 1.4 (holomorphy, Cauchy–Riemann equations)**.** We say that $f = u + iv$ is *holomorphic* in the variable z_ν if

$$\frac{\partial f}{\partial \bar{z}_\nu} = 0.$$

This is equivalent to the Cauchy–Riemann equations

$$\frac{\partial u}{\partial x_\nu} = \frac{\partial v}{\partial y_\nu}, \qquad \frac{\partial u}{\partial y_\nu} = -\frac{\partial v}{\partial x_\nu}.$$

Proof. By definition,

$$\frac{\partial f}{\partial \bar{z}_\nu} = \frac{1}{2}\Big(\frac{\partial}{\partial x_\nu} + i\frac{\partial}{\partial y_\nu}\Big)(u+iv) = \frac{1}{2}\Big[\Big(\frac{\partial u}{\partial x_\nu} - \frac{\partial v}{\partial y_\nu}\Big) + i\Big(\frac{\partial v}{\partial x_\nu} + \frac{\partial u}{\partial y_\nu}\Big)\Big].$$

A complex number is 0 exactly when its real and imaginary parts both vanish, so $\partial f/\partial \bar{z}_\nu = 0$ is equivalent to the Cauchy–Riemann equations above. ☐

Convergence of power series

Note 1.3. If, in a neighborhood of each point a of W, $f(z)$ can be expanded as a convergent power series

$$f(z) = \sum_{k_1,\dots,k_n \ge 0} c_{k_1\cdots k_n}(z_1 - a_1)^{k_1}\cdots(z_n - a_n)^{k_n},$$

then $f(z)$ is a holomorphic function.

Prop 1.1. *If the power series $\sum_{k_1,\dots,k_n\ge 0} c_{k_1\cdots k_n}(z_1 - a_1)^{k_1}\cdots(z_n - a_n)^{k_n}$ converges at some point w, then it converges absolutely for every z satisfying*

$$|z_\nu - a_\nu| < |w_\nu - a_\nu| \qquad (\nu = 1,\dots,n).$$

Proof. We may assume $a = 0$ (replace $z_\nu - a_\nu$ by z_ν). Since the series converges at w, its terms are bounded: there is a constant $C < \infty$ with

$$|c_{k_1\cdots k_n}\, w_1^{k_1}\cdots w_n^{k_n}| \le C \qquad (\forall\, k_1,\dots,k_n).$$

For $|z_\nu| < |w_\nu|$,

$$|c_{k_1\cdots k_n}\, z_1^{k_1}\cdots z_n^{k_n}| = |c_{k_1\cdots k_n}\, w_1^{k_1}\cdots w_n^{k_n}| \prod_{\nu=1}^{n}\left|\frac{z_\nu}{w_\nu}\right|^{k_\nu} \le C\prod_{\nu=1}^{n}\left|\frac{z_\nu}{w_\nu}\right|^{k_\nu},$$

and therefore

$$\sum_{k_1,\dots,k_n\ge 0} |c_{k_1\cdots k_n}\, z_1^{k_1}\cdots z_n^{k_n}| \le C\prod_{\nu=1}^{n}\sum_{k=0}^{\infty}\left|\frac{z_\nu}{w_\nu}\right|^{k} = C\prod_{\nu=1}^{n}\frac{1}{1 - |z_\nu/w_\nu|} < \infty.$$

Hence the series converges absolutely. ☐

Theorem 1: power-series expansion

Thm 1. *Let $W \subseteq \mathbb{C}^n$ be a domain. If $f(z)$ is continuous on W and holomorphic in each variable z_ν separately (the other variables being held fixed), then $f(z)$ is a holomorphic function of the n variables $z = (z_1, \ldots, z_n)$.*

Proof. Let a be an arbitrary point of W. Since W is open, we may choose $r = (r_1, \ldots, r_n)$ with

$$\{\, z \mid |z_\nu - a_\nu| \le r_\nu,\ \nu = 1, \ldots, n \,\} \subset W.$$

As f is holomorphic in z_1, the Cauchy integral formula in the z_1-plane gives

$$f(z_1, z_2, \ldots, z_n) = \frac{1}{2\pi i} \oint_{|w_1 - a_1| = r_1} \frac{f(w_1, z_2, \ldots, z_n)}{w_1 - z_1}\, dw_1.$$

Repeating this for each variable, and using that f is continuous so that the iterated integral equals the multiple integral (the order of integration may be exchanged), we obtain

$$f(z) = \frac{1}{(2\pi i)^n} \oint_{|w_1 - a_1| = r_1} \cdots \oint_{|w_n - a_n| = r_n} \frac{f(w_1, \ldots, w_n)}{(w_1 - z_1) \cdots (w_n - z_n)}\, dw_1 \cdots dw_n.$$

Now, noting that $|z_\nu - a_\nu| < r_\nu = |w_\nu - a_\nu|$, each factor expands as

$$\frac{1}{w_\nu - z_\nu} = \frac{1}{w_\nu - a_\nu} \cdot \frac{1}{1 - \dfrac{z_\nu - a_\nu}{w_\nu - a_\nu}} = \sum_{k=0}^{\infty} \frac{(z_\nu - a_\nu)^k}{(w_\nu - a_\nu)^{k+1}}.$$

Substituting these and integrating term by term,

$$f(z) = \sum_{k_1, \ldots, k_n \ge 0} c_{k_1 \cdots k_n} (z_1 - a_1)^{k_1} \cdots (z_n - a_n)^{k_n},$$

where

$$c_{k_1 \cdots k_n} = \frac{1}{(2\pi i)^n} \oint_{|w_1 - a_1| = r_1} \cdots \oint_{|w_n - a_n| = r_n} \frac{f(w_1, \ldots, w_n)}{(w_1 - a_1)^{k_1 + 1} \cdots (w_n - a_n)^{k_n + 1}}\, dw_1 \cdots dw_n.$$

Hence f is expanded as a convergent power series in a neighborhood of each point of W, and is therefore a holomorphic function of n variables. □